

Víctor M. Fonte Chavez

Full Stack Developer | Data Scientist

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SUMMARY

Full Stack Developer with a strong background in **Python, Django, FastAPI, and React**, complemented by experience in **data analysis and AI/ML**. I have contributed to the development of web applications and APIs, integrating relational and non-relational databases (**PostgreSQL, MySQL, MongoDB**). My academic and research experience includes work in computer vision and generative models, with a co-authored publication on *time-adaptive video frame interpolation based on residual diffusion*. Bilingual in **Spanish and English**.

TECHNICAL SKILLS

- **Frontend:** HTML, CSS, JavaScript, React (Hooks, Context), Redux, React Router, Tailwind CSS, basic animations/visualizations (Recharts / D3).
- **Backend:** FastAPI; Django / Django REST Framework; Auth (JWT, sessions); background tasks; testing (PyTest).
- **Databases:** PostgreSQL, Microsoft SQL Server, MongoDB; ORMs (SQLModel/SQLAlchemy/Django ORM); schema design, indexing, migrations (Alembic/Django Migrations).
- **Cloud & DevOps:** AWS (basics), Docker, Git/GitHub, CI/CD (GitHub Actions), Linux.
- **Security & Quality:** Input validation, HTTPS, CORS, logging/observability, linting/formatting, code reviews.
- **ML/AI (plus):** PyTorch, Generative Modeling (Transformers, Diffusers), RAG systems, MCP; Computer Vision, OpenCV; model serving via REST.
- **Languages:** Python, JavaScript/TypeScript, SQL; (R and C++ – working knowledge).

PROFESSIONAL EXPERIENCE

•Computer Vision Software Developer

Feb 2024 – Present

UnderCurrency SA de CV

Remote

- Built a full-stack web platform for banknote analysis: **React** (Hooks, context) front-end and **Django Rest Framework** back-end with REST endpoints.
- Designed scalable image-processing services and asynchronous jobs; improved throughput and reliability for video/image pipelines.
- Deployed models (YOLOv9, SAM, Lama Impainting, EasyOCR) behind APIs; packaged services with **Docker**; configured CI with GitHub Actions.

•Data Scientist (3-month project)

Sep 2024 – Dec 2024

Extract Transform and Exploration (EDT)

Remote

- Detected and resolved data inconsistencies; designed a metric for efficient resource management.
- Automated reports in Python; added API endpoints for data access.

•Junior Research Assistant

2022 – 2024

CIMAT – Center for Research in Mathematics

Guanajuato, Mexico

- Co-developed deep learning pipelines for **video frame interpolation** (PyTorch); reproducible experiments and result dashboards.
- Co-authored the paper *time-adaptive video frame interpolation based on residual diffusion* (arXiv: 2504.05402); engineered training/evaluation code and datasets.
- Exposed trained models via **PyTorch** and documented Frontend application deployed in HuggingFace, por example: (HuggingFace Space: Multi-Input-Res-Diffusion-VFI-Space).

•Profesor (Mathematics & Statistics)

Nov 2020 – Aug 2022

Universidad Central “Marta Abreu” de Las Villas

Cuba

- Taught Calculus, Probability, and Time Series Analysis; designed practical projects and assessments.

EDUCATION

- Master in Computer Science

Centro de Investigación en Matemáticas (CIMAT), Mexico

Aug 2022 – Jun 2024

CGPA: –
- Bachelor of Mathematics

Universidad Central “Marta Abreu” de Las Villas, Cuba

Aug 2016 – Sep 2020

GPA: –

PUBLICATIONS

- Victor Fonte, Jean Bernard Hayet, Claudia Esteves. *time-adaptive video frame interpolation based on residual diffusion*. arXiv preprint 2504.05402 (2025).

LANGUAGES

- SpanishNative
- EnglishFull professional proficiency (speaking, writing, presentations)